

Geothermal heating plant Schwerin: Realization of a cascaded large-scale heat pump system for the utilization of a medium-depth geothermal system

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ABSTRACT

Stadtwerke Schwerin (municipal utilities of Schwerin, Northeast Germany) operate a gas and steam turbine power plant whose waste heat is used in a district heating grid. The district heating grid is operated with a sliding flow temperature of 130 °C ... 80 °C and a return flow temperature of 55 °C. The minimum load in the consumer system is approx. 10 MW. In order to increase the share of renewable energies, Stadtwerke Schwerin decided to use geothermal heat to cover the base load supply of the district heating grid via a large, industrial-sized heat pump system. In addition, cogeneration units are installed to generate the drive power for both the heat pumps and the geothermal plant. Their waste heat will also be fed into the district heating grid, further increasing the efficiency of the entire system.

From the production well of the geothermal plant, the thermal water is produced using an ESP (electric submersible pump) with a maximum flow rate of 150 m³/h and a wellhead temperature of 55.5 °C.

A new technical building will be erected for the geothermal heating plant in the immediate vicinity of the production well. In addition to the electrical periphery of the production pump and the above-ground plant components of the thermal water circuit (heat exchanger, coarse filter), the technical building also contains the large-scale heat pump system.

The use of heat pumps makes it possible to cool down the thermal water to 20°C. Due to the large cooling of the thermal water, 5750 kW of the geothermal energy can be used as the source for the heat pump system. The heat pumps achieve a COP (coefficient of performance) of approx. 4.35 at a design flow temperature of 80°C. Accordingly, the heating capacity of the machines is approx. 7,450 kW. This high energy efficiency is achieved not only by the sophisticated machine technology but also by a cascade connection of the heat

pumps. The heat source and heat sink are routed in opposite directions. In this way, each heat pump operates with only a small temperature spread in its evaporator and condenser, and the temperature lift that each of the machines must provide is small.

In Schwerin, it is planned to connect four heat pumps in series. The synthetic refrigerant R1234Ze, which has only a very low global warming potential (GWP) of 7, is intended as the working fluid for the machines. Commissioning of the plant is scheduled for early 2023.

1. INTRODUCTION

On February 28, 2022, the Intergovernmental Panel on Climate Change (IPCC) published its latest report on climate change. In this report, it was pointed out that the conversion to a fossil-free energy supply within the next 15 years is absolutely essential in order to meet the climate targets set in the Paris climate protection agreement.

Stadtwerke Schwerin have set itself the goal of decarbonizing its district heating supply by using biofuels and geothermal energy. The first geothermal plant of Schwerin is currently being built in its district Lankow. The project started with a feasibility study in 2015. In 2019 and 2021 two very promising geothermal wells were drilled into an aquifer with "only" 55.5°C wellhead temperature.

Medium-depth geothermal energy in combination with large-scale heat pumps will be essential for achieving the goals agreed in the Paris Climate Agreement. The geothermal plant in Schwerin might serve as an example for other regions in Europe with geothermal resources of moderate reservoir temperatures.

2. GEOLOGICAL CHARACTERISATION

The exploited reservoir formation is the lower part of the Exter Formation. The Exter Formations were deposited in a deltaic complex and contain two Sandstone horizons, which are important geothermal reservoirs for the North German Basin. The depositional environment of the lower part of the

formation can be subdivided into distributary channels, levee/crevasse splay complexes, sheetsands and interdistributary bay facies (Franz et al., 2018). In terms of hydrothermal exploration, distributary channels and associated lateral levee/crevasse splay complexes are the most relevant morphological units. These morphological units can contain sandstone horizons with thicknesses greater 20 m. The reconstruction of the depositional environment and related facies associations/morphological units (see Wolfgramm et al., 2014, Franz et al., 2018) predicted a distributary channel in the subsurface of the urban area of Schwerin. The reservoir was developed by two wells, Gt S6 and Gt S7. The wells encountered a succession of fine- to medium grained sandstones. This succession can be subdivided into multiple stacked point-bar deposits, superimposed and underlaid by the deposits of the levee/crevasse splay complexes. Table 1 summarizes the main reservoir parameters of both wells.

Table 1 Summary of the reservoir parameters

Parameter	Gt S6	Gt S7
Reservoir depth (m TVD)	1245	1220
Reservoir temperature (°C)	56.8	55.5
Reservoir thickness (m)	49	42
Filter length (m)	30	39.4
Porosity _{mean} (%)	30.7	22.7
Core Permeability _{mean} (mD)	6.2	6.2
Test permeability (D)	6.2	6.3
Porductivity (l/(s·bar))	35	18

3. PLANT DESIGN

The above ground parts of the plant can be divided into two sections. i) the brine cycle for the production, handling and transportation of the brine and ii) the heat pump cycle. The two parts are connected by heat exchangers to protect the heat pumps from the

corrosive brine. A simplified overview of the brine cycle is given in Figure 1 and the heat pump cycle is shown in Figure 2.

The brine cycle fulfils all typical requirements for deep open-loop geothermal systems. At the wellhead, a small, container-like building is constructed. It contains only the equipment needed in close proximity to the wellhead, like valves and measurement devices. The brine is extracted with an electric submersible pump (ESP), reaching a maximum volumetric flow of 150 m³/h and a temperature of 55.5 °C at the wellhead. The salinity is approximately 145 g/l, mainly NaCl. It is then transported over an underground pipe to the heating plant facility adjacent to the drill site. Most of the equipment is installed here, including the heat pumps and the electrical facilities. First, the brine is run through coarse filters to protect the downstream equipment from particles that might have been mobilized in the reservoir. The brine then flows through the heat exchangers where it is cooled down to 20 °C and the heat is transferred to the heat pumps. The heat exchangers are designed for minimal temperature differences between the hot and cold side which increases the amount of heat that can be used and also the efficiency of the heat pumps as we will see later. The brine is then directed through another underground pipe to the injection site, approximately 850 m away.

There it is filtered again, this time with especially fine filters to protect the sandstone aquifer. The filters are placed in a building beside the drill site, so that there's no need to remove them in case of workover on the well. The wellhead of the injection site is also covered by a container-like building that only contains the equipment needed in close proximity. An injection pump is not needed due to the good properties of the aquifer.

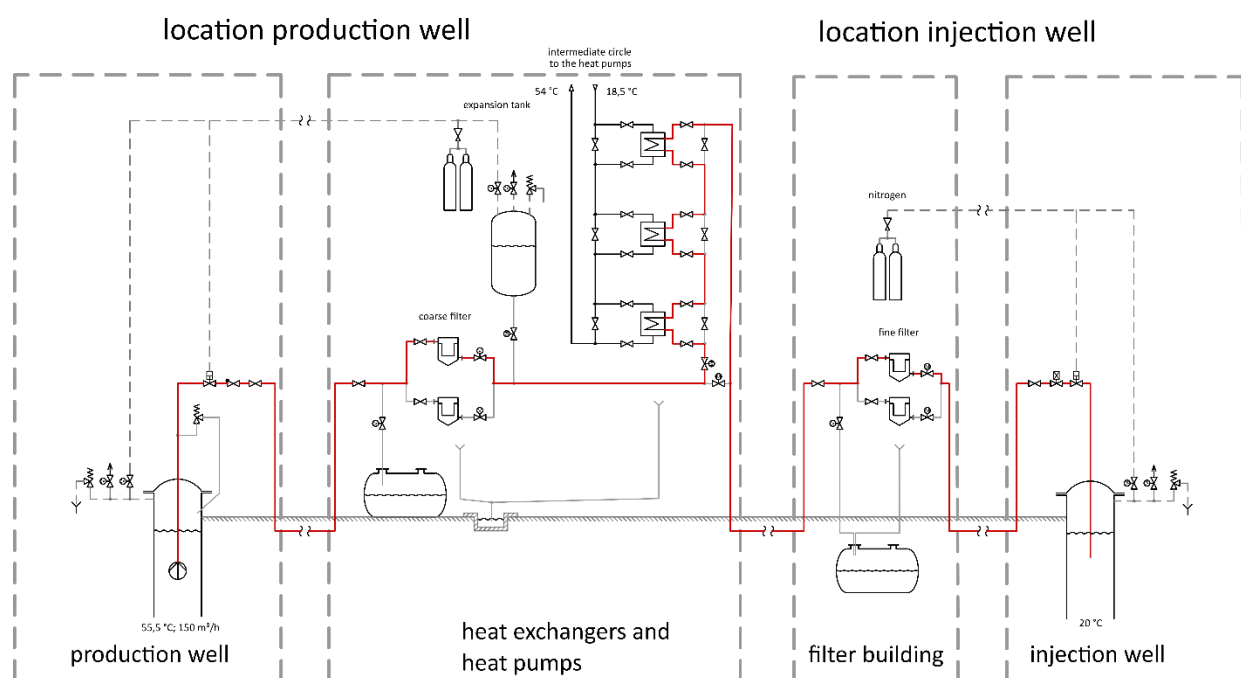


Figure 1: Brine cycle in Schwerin. The red line marks the path the brine takes through the plant.

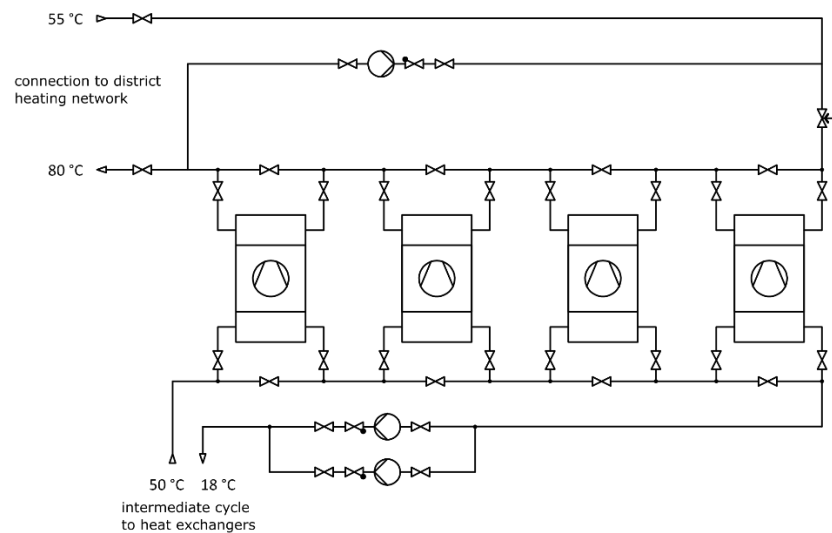


Figure 2: The heat pumps in Schwerin are connected in series, in opposite direction on the hot and cold side, to increase the overall efficiency of the plant.

The brine's high salinity of 145 g/l requires certain measures against precipitation, scaling, corrosion and the contamination of groundwater must be taken.

All equipment must have an adequate resistance against corrosion. Due to the relatively low temperatures, pipes and tanks can be made of Polypropylene (PP) or Polyethylene (PE) and most valves can be protected by membranes. Where this is not possible, high grade stainless steel must be used. For the very thin plates in the heat exchangers, only titan is suitable.

To avoid precipitation, scaling and increased corrosiveness, the brine needs to be put under a pressure of at least 2 bar at any time and the intrusion of oxygen must be avoided. The pressure is preserved by a valve right before the injection well. It increases the ESP's total dynamic head until the necessary pressure is reached in all parts of the plant. During downtime the pressure is preserved by an expansion tank with a nitrogen cushion. Additional nitrogen cushions are created in the annulus spaces off the wellheads.

The specific mineralisation of the brine leads to the classification as "hazardous to water" under German law. All parts filled with brine must be equipped with a second barrier, such as double walls or a basin, to avoid the leakage of brine. Inside the buildings, this is achieved easily by installing a watertight floor. Special, double walled equipment must be used outside of the heating plant. Additionally, sensors are used to automatically detect and react to a leakage.

The heat pumps are placed in the technical building. To match the exact heat demand and to reduce investment costs, four standard units are used. They are connected to the heat exchangers with an intermediate, water-containing cycle. The hot side is directly connected to the district heating network of Schwerin. The geothermal plant is designed to provide the base load. It is connected parallel to the other heaters. Most of the time, the heat demand is larger than what the heat pumps can provide and they can run at full

load. The heat pumps can provide a maximum temperature of 83 °C. It is possible to rise the temperature with other heaters afterwards if the heating network requires temperatures above that threshold.

In the design phase of the plant, it's necessary to find a heat pump which is fitted for the temperatures of both the source (geothermal reservoir) and the heat sink (the district heating network). In Schwerin, machines with shell and tube heat exchangers, screw compressors and R1234Ze as refrigerant were chosen. In the decision for the refrigerant, its low global warming potential (GWP) of seven played an important role. The main alternative, ammonia, was rejected for its toxicity.

Electricity is mainly needed for the heat pumps and the ESP. To lower the overall energy costs, two cogeneration units are installed in a separate building. The cogeneration units are strictly driven by the immediate electricity demand of the geothermal plant. They use natural gas to operate and their waste heat is fed to the district heating network.

4. CASCADE DESIGN OF LARGE SCALE HEAT PUMPS

The efficiency of a heat pump is expressed in the coefficient of performance (COP). It is defined as the quotient of the heat output and the used drive power.

$$COP = \frac{Q_{out}}{P_{in}} \quad [1]$$

The higher the COP, the fewer electric power is needed to extract a given amount of heat from the source.

The highest physically possible COP for a heat pump working between the highest temperature of the heat sink T_H (supply temperature of the district heating network) and the lowest temperature of the source T_L (injection temperature of the brine) is defined by the Carnot Cycle.

$$COP_{Carnot} = \frac{T_H}{T_H - T_L} \quad [2]$$

Due to imperfections and losses, this ideal process can never be exactly recreated. However, the conditions defining the COP_{Carnot} still have a big impact on the real COP, which makes their optimisation the first step to increase the efficiency of a heat pump system.

A big temperature difference between source and sink results in a low COP. As shown in formula [2] only the output temperatures of source and sink are relevant. In this project, source and sink have a considerable temperature lift, that can be distributed over multiple heat pumps by connecting the output of one machine to the input of the next one. This way, for all heat pumps but the last one, the highest temperature is reduced and therefore the COP increased. If the cold side is connected in the same way but with the flow in the opposite direction, the temperature difference can be reduced further, except for the first heat pump.

In Figure 3 a single heat pump is compared with a system of three machines connected in the way described above. Assuming an equal heat output for all machines

$$Q_{hp1} = Q_{hp2} = Q_{hp3} = Q_{plant}/3, \quad [3]$$

the theoretically best COP can be increased from

$$COP_{Carnot} = \frac{(80 + 273.15)K}{(80 - 20)K} = 5.89 \quad [4]$$

to

$$\begin{aligned} COP_{Carnot} &= \frac{1}{Q_{plant}} \left(\frac{(80 + 273.15) K}{(80 - 40) K} Q_{hp1} + \right. \\ &\quad \left. \frac{(70 + 273.15) K}{(70 - 30) K} Q_{hp2} + \frac{(60 + 273.15) K}{(60 - 20) K} Q_{hp3} \right) \\ &= \frac{(8.83 + 8.58 + 8.33)}{3} = 8.58 \end{aligned} \quad [5]$$

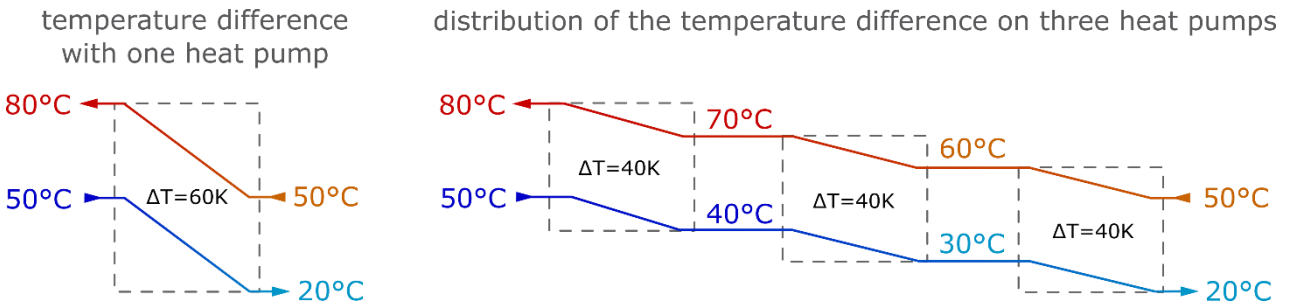


Figure 3: Increasing the overall COP by distributing the temperature difference over multiple heat pumps.

This increase also affects the real obtainable COPs considering all the losses compared to the ideal cycle.

Multiple heat pumps not only optimize the COP but also increase overall investment costs for the machines and the other technical equipment. There is an economical optimum for every project. In the case of Schwerin, it was reached with four heat pumps.

The plant in Schwerin, designed with the principles described above, produces 6.9 MW of heat at full load with temperatures of 80 °C / 55 °C on the hot and 55.5 °C / 20 °C on the cold side. During one year, it will use 1.1 Mio m³ of brine and 14.4 GWh electric energy to produce 60 GWh of heat. This results in a mean COP of 4.2.

The municipal utilities of Schwerin will supply up to 10 % of the district heating grid's demand with climate neutral geothermal energy and will achieve this goal economically viable through the use of new, optimised heat pumps. This makes the example of Schwerin relevant for all regions in Germany and Europe with similar geothermal recourses and moderate reservoir temperatures.

5. EFFICIENCY EVALUATION OF A GEOTHERMAL PLANT USING THE COP

Large-scale heat pumps have the potential to cover a large part of the heat supply in urban areas and are 100% fossil free if combined with renewably produced electricity. By combining geothermal reservoirs of moderate temperatures with large-scale industrial heat pump systems, a large number of additional geothermal resources can now be used for urban heat supply.

The use of brine from a medium-depth geothermal plant as a source of environmental energy increases the COP of the overall plant and simultaneously reduces the required circulation volume compared to an alternative source such as air, groundwater or seawater to an even greater extent. In urban areas in particular, the available space is limited, so that the circulation volume of the source must be as small as possible.

For each heat pump-driven heating plant, a total COP can be calculated regardless of the source. This COP includes not only the electricity needed for the heat pumps, but also the electrical energy needed to circulate the source.

$$COP_{total} = \frac{Q_{out}}{P_{HeatPump} + P_{Circulation} + P_{other}} \quad [6]$$

Even a geothermal heating plant which is operated in direct use without heat pumps, which means that the power of the heat pump is zero, can be characterized in the same way. Geothermal plants in direct heat exchange achieve a COP of up to 35, depending on the geological reservoir. The annual performance factor can also be determined on the basis of the real annual electrical input.

This relationship is shown under idealized and normalized conditions in Figure 4, using the formula [6]. The achievable COP of geothermal heating systems in direct heat exchange, with heat pumps and heat pump systems with other sources were estimated. With rising source temperatures the COP increases and the volumetric flow decreases.

The COPs for the heat pump systems were estimated based on a cascade circuit with 4 heat pumps and considering real machine data. In order to make all heating systems comparable, it was assumed, that an infinite heating network with a supply temperature of 80 °C and a return temperature of 55 °C has to be supplied. The heat pump systems were normed to 10 MW heating capacity.

Due to a large temperature drop of thermal water, such as in Schwerin to 20 °C, the nominal COP of a medium-depth geothermal system with heat pumps is only slightly better than the COP of a groundwater or seawater heat pump system. This results in a reduction of electrical power by 20 % to 30 %. However, the required circulation volume is much smaller, since the large temperature spread increases the heat that can be extracted from a certain amount of thermal water. River heat pumps need an immense volumetric flow because the temperature drop is limited to about 5 °C. At many locations, therefore, it is not possible to use river or ground water on a large scale. The same is the case for air-source heat pumps, which would have to circulate a large amount of ambient air to generate megawatts of power.

When evaluating fuel-free heat generation systems, the COP can always be used as an evaluation parameter. This is especially true for geothermal heating systems. In connection with the quantification of the required circulation flow, these two parameters represent the most important plant parameters for the potential assessment in the design process of fossil-free heat generation plants.

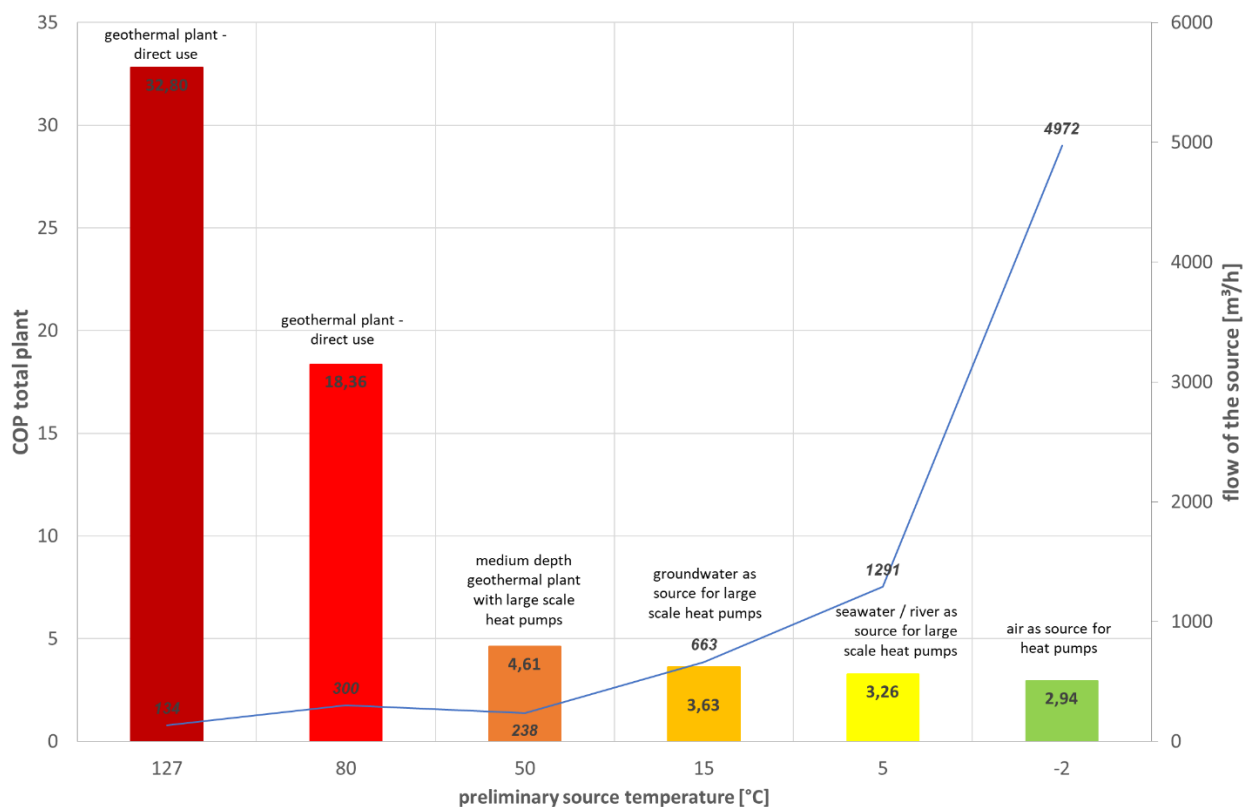


Figure 4: COP and source flow comparison of different sources which can be used for large scale heatpumps. The heat output is normalized to 10 MW feeding heat into an 80 °C to 55 °C heating net. red column: geothermal plant direct use (e.g. Munich Area); medium red column: geothermal plant direct use (Neustadt Glewe); dark orange cloumn: medium depth geothermal plant with large scale heat pumps (Schwerin); orange cloumn: groundwater as source (Mannheim); yellow column: seawater / riverwater as source; green column: air as source

6. CONCLUSIONS

When completed in 2023, the Schwerin-Lankow geothermal project will be the first medium-depth geothermal plant in Germany to use a large-scale heat pump system to provide a heating capacity of more than 5 MW for an urban district heating network. The use of a cascade connection of large-scale heat pumps, which have made an enormous technical and economic development in the last five years, makes it possible to use an extremely productive aquifer, even though it only has a wellhead temperature of approx. 55 °C. This means that even with low wellhead temperatures, it is still possible to produce heat from a medium-depth geothermal system.

Despite the salinity of the thermal water, the low production temperature allows that cost-intensive stainless-steel materials can mostly be avoided in the geothermal loop, which significantly improves the overall economy and reduces dependence on expensive raw materials. Taking into account the market situation and applicable subsidy guidelines in 2020, the Schwerin geothermal plant was already more economical than a CHP plant.

The use of sandstone aquifers for geothermal plants has been little used due to the complex requirements for plant design and the rather moderate achievable heat outputs to date. In view of the changed market situation for fossil fuels in 2022 and taking into account the technical advances in large-scale heat pumps, the previous reservations are obsolete.

The conceptual approach of a geothermal plant with large-scale heat pumps pursued in Schwerin can be transferred to all shallow and medium-depth aquifers throughout Europe. The extensive use of geothermal energy is the energetically best and usually also most economical option for the necessary decarbonization of the heating sector.

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